Homework #8 (30 points) **Due:** Mar. 25, 2026, Wednesday

Purpose: This homework is to familiarize you with SQL and ORACLE SQL-Plus Environment.

1. Create the company database (for the schema details and data, see the file '*Company-DB-tables-info.pdf*') and populate it with the data provided in the Oracle SQL PLUS environment.

2. Your SQL commands should be placed in a file and run from the file
   - one file for creating tables, another file for inserting tuples, and another one for queries.
   - Use Microsoft .NET editor or Notepad++ to create and edit your files.
   - Use 'sql' suffix for file names (i.e., hw8-queries.sql).

3. In creating the tables, make sure to specify referential integrities as well as primary key constraints.

4. Apply check constraints for Salary attribute (20,000<=salary<=100,000) and Sex attribute (validate it for 'M' or 'F').

5. Write and execute the following test queries and keep track of the results.
   a. The first name, last name of employees who work at department 5.
   b. The first name, last name of every employee and name of his/her department.
   c. The first name, last name of employees who work at the 'Research' department.
   d. The first name, last name of employee who is the manager of the 'Research' department.
   e. The first name, last name of employees who work on the 'Computerization' project.

No blank lines inside CREATE TABLE commands (Oracle SQLPlus is sensitive to blank lines in CREATE TABLE, gives a syntax error).
As you have seen in homework #7, the order of creation of tables and insertion into tables makes a difference. You need to pay attention the referential integrity constraints and determine the correct create and insert order. There is one circular reference you need to pay attention. Employee.dno is a reference for Department table and Department.mgrssn is a reference to Employee table. *Oracle gives an error if you make a reference to a table that has not been created yet.*

Solution: Create Employee table before Department table and keep the foreign key constraint mgrssn referencing Employee. In creating Employee table, do not place the foreign key constraint dno referencing Department, defer it, add the following command after creating the department table:

alter table employee add `constraint EMPDEPTFK` *foreign key (dno) references department (dnumber)* `ON DELETE SET NULL;`

Use 'drop table' command before every create table command, and commit' command after every create table command, and after every insert group per table (also 'delete from' command before every insert group per table). . No blank lines inside CREATE TABLE commands (Oracle SQLPlus is sensitive to blank lines in CREATE TABLE, gives a syntax error). See the partial solution provided.

Partial solution:
`K:\Courses\2026-Spring\ITC341-ugur1a\_Global-Materials_\hw8-partial\`
Also, in Blackboard. Make copies of the files, edit each one, and add the rest of your solution.

How to put the screenlog in the file 'U:\itc341\hw8\hw8-out.txt' (after all testing is done; for final output)

(You should already have '**itc341**' folder in your U drive). Then, in U:\itc341, create a folder named **hw8**. Place all your files in **U:\itc341\hw8**. You also need to put your name, ITC 341 assignment number on top of every file as comments.

The example below is for create table commands file **`cr-company.sql`** only.

```
-- cr-company.sql
-- Creates the database schema for the COMPANY database.
-- Jim Goodwill
-- ITC 341 Homework 8     Mar. 20, 2026
```

Necessary SQL-Plus commands you need to use for the final run:

SQL> set linesize 150 -- needed for displaying larger tables
SQL> set echo on -- **this is a MUST** (for including commands results in the output file)
SQL> set pagesize 120 -- for a better display

SQL> spool U:\itc341\hw8\hw8-out.txt

-- run the commands from your files, one by one, in the following order

SQL> start U:\itc341\hw8\cr-company.sql -- assuming create table commands are in
-- U:\itc341\hw8\ '**cr-company.sql**' file
SQL> start U:\itc341\hw8\ins-company.sql -- assuming insert table commands are in
-- U:\itc341\hw8\ '**ins-company.sql**' file
SQL> start U:\itc341\hw8\q-company.sql -- assuming queries are in
-- U:\itc341\hw8\ '**q-company.sql**' file
SQL> spool off

-- The run results would be in the file 'U:\itc341\hw8\hw8-out.txt'.

What to hand in:

1. First, make sure the results are correct (i.e., testing needs to be completed first).
2. **Submit the entire hw8 folder.** Copy the entire hw8 folder into **K:\Courses\2026-Spring\ITC341-ugur1a\22482905\yourCMUglobalId** folder.
3. Zip your hw8 folder as **yourCMUid–hw8.zip** and submit to the Blackboard as well.